Mohammadreza Nemati

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Experience

Senior AI Engineer — 05/2025 - Present
Volkswagen Group of America — *Belmont, CA*

- Integrated a diffusion-based world-model approach into the multi-agent automotive HMI testing framework, predicting future environment states from candidate actions to enable more reliable long-horizon decision-making.
- Fine-tuned multimodal foundation models using SFT, DPO, and GRPO for planner, reasoner, grounder, verifier, and error-recovery components in real-device automotive HMI software testing.
- Compressed Qwen2.5-VL for autonomous-driving trajectory prediction, reducing parameters from 3.75B to 1.4B through pruning, then distilling knowledge from the 7B teacher model and applying quantization-aware training to preserve performance under constrained edge deployment.
- Built multimodal GUI-agent system for real-device Android HMI software testing, combining screen understanding, element grounding, tool-use actions, action verification, and error recovery across in-vehicle app workflows.
- Served custom VLM checkpoints with vLLM continuous batching and optimized attention backends to improve throughput and latency for agentic AI inference workloads.

Artificial Intelligence Research Assistant — 08/2022 - 06/2025
Case Western Reserve University — *Cleveland, Ohio*

- Optimized LLM inference memory by analyzing KV-cache key/value norm imbalance, achieving 2× memory reduction with >94% accuracy retention across models up to 70B parameters [paper link]
- Developed LIT-LVM using latent variable embeddings, structured regularization, and GNN-based higher-order interactions, outperforming GNN, Factorization Machine, and elastic net baselines in ranking and recommendation tasks [paper link]
- Developed DEASurv, a deep multimodal model leveraging entity-specific embeddings, attention mechanisms, and contrastive loss to improve prediction on large-scale medical datasets

Software Engineer - AI/ML Intern — 02/2024 - 07/2024
Sherwin-Williams — *Cleveland, Ohio*

- Designed and deployed real-time vision-transformer inference pipeline for SAM, tuning latency and improving segmentation quality by 7% IoU through inference evaluation and pipeline optimization.
- Built VLM-based visual grounding system integrating LLaVA with vision transformers, implementing cross-modal alignment for natural-language object grounding and segmentation.

Data Science Research Intern — 05/2023 - 09/2023
Beyond Limits AI — *Glendale, California*

- Developed and deployed Graph Neural Network models (GCN, GAT, GraphSAGE) using PyTorch Geometric for graph-structured data, achieving 10% accuracy improvement over baseline approaches
- Fine-tuned Llama-2 using PyTorch distributed training with FSDP/DDP for multi-GPU optimization and quantization for efficient deployment.

Skills

Agentic AI: GUI Agents, Long-Horizon Planning, Task Decomposition, Action Verification, Error Recovery

Foundation Models / GenAI: Python, C++, PyTorch, Diffusion Models, World Models, Vision-Language Models, Multimodal Reasoning, Prompting

Evaluation / Optimization: Model Evaluation, Quantization, Knowledge Distillation, Pruning

Deployment / Frameworks: vLLM, CUDA, Distributed Training (FSDP/DDP)

Education

Case Western Reserve University – PhD, Computer Science — Cleveland, OH | 2025

University of Toledo – MS, Computer Science — Toledo, OH | 2020

Shiraz University – BS, Electrical Engineering — Shiraz, Iran | 2017